Multi-Information-Enhanced Knowledge Embedding in Hyperbolic Space

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Abstract. Knowledge Graph Representation Learning(KGRL) aims to map entities and relationships into a low-dimensional dense vector space. Most of the existing models focus only on the information of the triple when doing representation learning, ignoring the rich external semantic information. At the same time, these models consider entities and relations as static and single representations, so the knowledge represent ability is poor. Accordingly, we propose a novel knowledge graph representation model which enhanced knowledge graph embedding with multi-information. Firstly, our model carries out text enhancement and hyperbolic space embedding of triples in the knowledge graph respectively; Secondly, we concatenate the enhanced vector. Then, the concatenated vector through two transformation layer to fuse the semantic information and spacial information. Finally, we use the fused information to learn the context information through the Transformer coding layer, which will dynamically produce the final representation of the entity based on its context. Experimental results show that our model has a great improvement over other models. In the link prediction task, the evaluation protocol Hits@10 and MRR in the public dataset FB15k improve by 28.4% and 29.5% compared with the translation model. Compared with state-of-the-art model, the improvement is 2.5%, 6.3%.

Keywords: Knowledge Graph · Representation learning · Transformer

1 Introduction

Knowledge Graph is an important technique for structuring knowledge. To store and utilize structured knowledge efficiently, Knowledge Graph such as: WordNet (Miller 1995) [13], DBpedia (Auer et al. 2007) [1], Freebase (Bollacker et al. 2008) [2] and other classical Knowledge Graph are constructed by combining expert annotation and computer automatic annotation. Knowledge Graph store huge numbers of structured data in the form of triples. These triples are usually represented as (h, r, t), where h represents the head entity, t represents the tail entity, and r represents the relation.

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Large-scale knowledge graph has problems of low computational efficiency and sparse data. In addition, knowledge graph completion still far away from complete due to the constantly emerging new knowledge. At present, more and more scholars aim to project knowledge graph into low-dimensional and continuous vector space, so as to improve the computational efficiency of knowledge graph and alleviate the problem of data sparsity. These representation models could be divided into three types, Translation Models [3,9,11,21], Neural Network Models [4,10] and Tensor Decomposition Models [6,19,25]. Among them, translation model is the most classical method, which has received a large number of attention and application because it only needs fewer parameters in training and achieves better knowledge expression effect. Though these methods have achieved excellent performance, they ignore the rich external semantic information as well as the contextual information of the knowledge graph structure. In this way, some methods using these external semantic information become an active research for knowledge graph completion. Ruobing Xie [24] proposed a knowledge representation learning method TKRL based on entity hierarchy type embedding, and Wang [22] proposed TEKE for knowledge enhancement by using text description information of entities. In addition to considering the description information of entities, some scholars consider the information of the entity's neighbour nodes. The survey of Provenance-Aware Knowledge Representation [16] also mentioned the improvement of context to knowledge graph representation learning.

All the above models are modeled in Euclidean space. However, due to geometric constraints, these models often require high dimensions for knowledge representation. Some scholars begin to explore different Spaces for modeling. ManifoldE [23] maps entities and relations to manifold space. Kolyvakis et al. [7] proposed HyperKG, which embed entities and relations into hyperbolic space, and use the structural features of hyperbolic space to capture the hidden information of data to improve knowledge representation ability. In the latest research, Sun et al. [18]. Proposed the RotatE model, which represents entities into complex space and relations into rotation translation. However, the RotatE model does not take into account the context information of graphs when encoding, making them ineffective in dealing with complex relations.

According to the issue mentioned above, we propose a novel model to embed knowledge graph with multi-information. Inspired by multi-modal information fusion, we propose an effective method to fuse external semantic information and spatial structure information into the same space. At the same time, we take the context entity information into the account and consider the training process of the fact triplet as a Seq2Seq process, so that entities and relations are not a single static model.

We summarize the main contributions of our work as follows:

- We propose a novel method for information fusion of external semantic information and hyperbolic space information, which strengthens the constraint of context information learning.

- Instead of the previous methods that allow a single static representation for each entity or relations, we treat triple, entity and relation as a sequence and a context respectively, which is good for handling complex relationships.
- Extensive experiments have done to show our superiority in dealing with complex relationships. Compared with the state-of-art models, our model has significant advantages.

2 Related Work

Representation Learning of Knowledge Graph is to represent entities and relations in knowledge base as dense low-dimensional vectors. The classic model is the translation model, Bordes et al. [3] regard the process in which the head entity relates to the tail entity through relations as the translation process, and then measure the rationality of each triplet with a score function. After that, a series of translation models are proposed [11,21]. Another is the tensor factorization methods such as DisMult [25] and SimplE [6]. While these methods have few parameters and easy to model, they ignore the structural information of the knowledge graph itself and the abundant external corpus information which fail to improve the ability of dealing with complex relationships. In addition to some of the models mentioned above, many scholars have proposed models integrating external information. The first category is the research of embedding space. Many scholars believe that Euclidean space is not suitable for knowledge representation modeling due to geometric limitations. They try to transform the representation space into manifold space [23], hyperbolic space [7], complex space [18], and the latest research maps entities and relations in the knowledge graph to Quaternion space. The second category is the research on external semantic information, which integrates the information of entity type and entity description information into the representation of entity to improve the representation ability of entity. The third category is the study of knowledge graph structure. Quan Wang et al. proposed a contextualized knowledge graph embedding model (COKE) [20], arguing that the representation of entities and relations should change with context. They utilize the transformer to exploit the contextualized and dynamic representations for entities and relations. Besides Liu et al. [8,12] also used Knowledge graph to enhance the item in recommend system,arguing that they use attention-enhanced dynamic convolutional network to enhance the item. Based on Coke's and attention-enhanced idea, our paper further uses attention mechanism to integrate external entity description information to eliminate semantic deviation in multi-dimensional space, improve the ability to deal with complex relations.

3 Preliminary

In this section, we will introduce some important concepts covered in this article, including the definition of knowledge graphs and the concept of Hyperbolic Space. Finally we will give the definition of our work.

3.1 Knowledge Graph

Knowledge Graph is defined as a directed graph used to store structured information about real-world entities and facts. In this paper, given a Knowledge Graph $\boldsymbol{G} = \{(h, r, t)\} \subset \mathrm{E} \times \mathbb{R} \times \mathrm{E}$, where E and $\mathbb{R}$ denote the entity set and relation set respectively. A fact triplet (h, r, t) describes a head entity $h \in \mathrm{E}$ connected to a tail entity $t \in \mathrm{E}$ through a relationship $r \in \mathbb{R}$. After knowledge graph representation learning, the head vector is represented by $\boldsymbol{h}$, the relation vector by $\boldsymbol{r}$, and the tail entity vector by $\boldsymbol{h}$.

3.2 Hyperbolic Geometry

Here we briefly give some concepts of Hyperbolic Space geometry, a more detailed explanation can be found in review article [14]. Hyperbolic space has the ability to reflect data hierarchy and has a large space capacity. In this paper, we choose the Poincaré-ball model, which has the feasibility of gradient optimization in this task. Here we define a Poincaré sphere by a Riemannian manifold $\left(\mathrm{B}^d, g_x\right)$, where $B^d = \left\{x \in \mathbb{R}^d, \|x\| < 1\right\}$ is a d-dimensional Poincaré ball and g_x is the Riemannian metric tensor. For two points $\boldsymbol{u}$ and $\boldsymbol{v}$ in hyperbolic space, their distance function can be expressed as:

$$d(\boldsymbol{u}, \boldsymbol{v}) = \operatorname{arcosh}\left(1 + 2\frac{\|\boldsymbol{u} - \boldsymbol{v}\|^2}{(1 - \|\boldsymbol{u}\|^2)(1 - \|\boldsymbol{v}\|^2)}\right) \tag{1}$$

This formula is symmetric, and as nodes move from the origin towards the poincaré ball boundary, their distance will increase exponentially. This will provide more space for representation learning of entities.

Vector translation in poincaré ball model is also different from in Euclidean space. It is generally defined by Mobius addition, and its formula is as follows:

$$\boldsymbol{u} \oplus \boldsymbol{v} = \frac{\left(1 + 2\langle \boldsymbol{u}, \boldsymbol{v}\rangle + \|\boldsymbol{v}\|^2\right)\boldsymbol{u} + \left(1 - \|\boldsymbol{u}\|^2\right)\boldsymbol{v}}{1 + 2\langle \boldsymbol{u}, \boldsymbol{v}\rangle + \|\boldsymbol{v}\|^2\|\boldsymbol{u}\|^2} \tag{2}$$

3.3 Task Definition

The task of knowledge graph completion is to predict unknown nodes based on existing nodes and relations in the knowledge graph $\boldsymbol{G}$. For example, predicting the tail entity with the given entities and relation$(h, r, ?)$. Specifically, the task is to design a scoring function that assign a higher score for positive sample triple than the negative sample triple.

4 Our Method

The goal of this work is to learn a model that can encode complex relations such as 1-N, N-1 and N-N. As depicted in Fig. 1, our model contains four key parts:

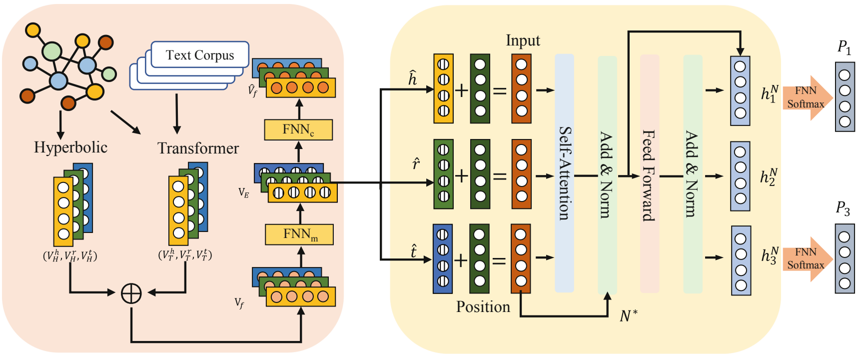


Fig. 1. An architecture overview of our model. The framework of the model contains four key modules: (1)Textual Embedding; (2)Hyperbolic Translation; (3)Information Fusion Network; (4)Contextualized Embedding

(1) Textual Embedding: Based on our approach in previous papers, we use external descriptive text for semantic enhancement of entities through encoders.

(2)Hyperbolic Translation: We translate the fact triplet of the knowledge graph into hyperbolic space and this will be the spatial information as the part of information fusion.

(3) Information Fusion Network: We extract features by maximizing the correlation between inputs from multiple information sources.

(4) Contextualized Embedding: For entities and relations after information fusion, we use Transformer to encode and obtain contextual representations of entities and relationships.

4.1 Textual Context Embedding

In order to associate entities in the knowledge graph with external text corpus, we employ an entity linking tool AIDA for entity annotation. Referring to the co-occurrence network used by TEKE [22], we obtain a unique description text for each entity. Given a knowledge graph $\boldsymbol{G}$ and an external text corpus $T = \{S_1, S_2, \cdots, S_n\}$, where S_i represents the text statements in the external text corpus. We represent the input sequence as $S = (w_0, w_2, \cdots, w_n)$, where w_0 represents the [CLS] tag and the other elements represent words in the text statement. For each word in the input, we pre-process the position information to the representation of each word as follows:

$$\mathbf{w}_i^0 = \mathbf{w}_i^{ele} + \mathbf{w}_i^{pos} \tag{3}$$

The main function of position coding is to identify the position of elements in a sequence. We stack multiple Transformer encoders into coding layers and use the output of each layer as the input of the next layer of encoders. The encoding formula is as follows:

$$\mathbf{w}_i^l = \text{ Transformer} - \text{Encoder}\left(\mathbf{w}_i^{l-1}\right) \quad l = 1, 2, \cdots, L \tag{4}$$

where $\boldsymbol{w}_i^l$ denotes the output of L-th layer of encoder. We take the [CLS] tag representation as the final representation of the entity. The representation of the entity has learned information about the description text. We define the text enhanced entity as V_T.

$$\boldsymbol{V_T} = \mathbf{w}_0^L \tag{5}$$

4.2 Hyperbolic Translation

Given a triple of a knowledge graph, the approach of translation models usually treats the relationship as a translation from head entity to tail entity. The biggest problem with these translation models in Euclidean space is that they require high dimensions to achieve adequate representation of entities. The spatial properties of hyperbolic space make it more accurate to express the data of hierarchy and topological structure. Hyperbolic space is curved, and the carrying capacity of the space increases exponentially with the distance from the center. Hence, vectors that require higher dimensions in Euclidean space can be simply represented in lower dimensions in hyperbolic space. The related geometric theory of hyperbolic space can be found in Sect. 3.2.

Based on previous work in this paper, we use hyperbolic relation-specific transformations on entity representations and this will lead to high complexity overhead. In our exploration of hyperbolic space, we refer to the translation method used in HyperKA [17] and treat entity relations as translations like TransE, avoiding complex overhead. Therefore, our distance function is defined for a triple $\nu = (h, r, t)$:

$$S(\nu) = d\left(\boldsymbol{V_H^h} \oplus \boldsymbol{V_H^t}, \boldsymbol{V_H^r}\right) \tag{6}$$

where V_H^h,V_H^r,V_H^t denote the embedding for h, r, t respectively. We minimize the following constrastive learning loss:

$$\text{Loss } = \sum_{(\nu)\in G} \sum_{(\nu')\in G^-} \left[\gamma + S(\nu) - S(\nu')\right]_+ \tag{7}$$

where ν' denotes the randomly generated negative sample training data in knowledge graph. γ is the margin and we hope $S(\nu')$ ¿ γ.

Finally, we define the vector after hyperbolic TransE as V_H.

4.3 Information Fusion

Our previous work uses relations-specific hyperbolic transformation on the text-enhanced representation of triples in the knowledge graph, which achieves better representation than text-enhanced and hyperbolic representation alone. However experimental results show that this leads to high complexity overhead. Inspired by Multi-modal Machine Learning, we obtain the enhanced triplet $\left(V_T^h, V_T^r, V_T^t\right)$ and the hyperbolic space mapped triplet $\left(V_H^h, V_H^r, V_H^t\right)$ through text enhancement and hyperbolic TransE respectively. We consider these two triples from different sources or forms of information as two modals. Due to different modals show different ways and view things from different perspectives, there will be some crossover in the fusion and process, which will lead to redundancy of information. But this also brings complementary features that are more expressive than a single feature. The text enhanced entity has the description information of the entity itself, and the semantic information is more complete. Entities after Hyperbolic TransE pay more attention to the hierarchical structure of the entity itself, making it more expressive. Inspired by our previous work and the traditional feature fusion algorithm [26–28], we propose a novel method to fuse the two different representations of fact triples in the knowledge graph. We consider enhanced triplet $\left(V_T^h, V_T^r, V_T^t\right)$ and hyperbolic TransE triplet $\left(V_H^h, V_H^r, V_H^t\right)$ as the input. Then we first concatenate the entity and relation separately to obtain a triple $\left(V_f^h, V_f^r, V_f^t\right)$. Then we reduce the dimension of head entity, relation entity and tail entity respectively through a FNN layer to $\left(V_E^h, V_E^r, V_E^t\right)$. Then, we use this to reconstruct the originally concatenated triple $\left(\hat{V}_f^h, \hat{V}_f^r, \hat{V}_f^t\right)$. We use two Feed-forward Neutral Network as the transformation layer to produce the fused vector. Referring to [15], we used the mean square error as the loss function of multi-modal feature fusion. Taking the head entity as an example, the MSE loss function is as follows:

$$Loss_f = \left\|\hat{V}_f^h - V_f^h\right\|^2 \tag{8}$$

Through the Information Fusion part, we consider the middle triple$\left(V_E^h, V_E^r, V_E^t\right)$ as the information fusion representation. In section4.4 we convert $\left(V_E^h, V_E^r, V_E^t\right)$ to $\left(\hat{h}, \hat{r}, \hat{t}\right)$ for symbolic simplicity.

4.4 Contextualized Embedding

Context information can help entities generate their representations dynamically, making them better able to handle complex relationships. Inspired by Quan Wang et al., we refer to the most basic form of graph context in their model to learn about contextual information embedding. We learn context information by masked multi-head self-attention. The idea of Transformer and Bert is well known to most people and the specific formulas are not shown here. The more detail of formula can check the paper [5].

Since we get the information fusion triple $\left(\hat{h}, \hat{r}, \hat{t}\right)$, we see this triple as the input element of the Transformer. For each input triple we add the position embedding and then obtain the final input:

$$h_i^0 = \hat{h} + \hat{h}^{pos} \tag{9}$$

After through the Transformer Decoder, we obtain:

$$\hat{h_i^N} = \text{Transformer}\left(\hat{h}_i^{N-1}\right), \quad N = 1, 2, \cdots, N, \tag{10}$$

During training, inspired by Coke [20], we create two training instances. For a triple (h, r, t), one is to replace t and the other is to replace h, then we put this triple into the Transformer encoder to generate a new sequence $\left(\hat{h}_1^n, \hat{h}_2^n, ...\hat{h}_n^N\right)$. After a feedforward layer and a standard softmax classification layer, we get the target predicted entity:

$$\mathbf{P}_1 = \text{softmax}\left(\mathbf{W}f\left(h_1^N\right)\right) \tag{11}$$

$$\mathbf{P}_n = \text{softmax}\left(\mathbf{W}f\left(h_n^N\right)\right) \tag{12}$$

where $\mathrm{W} \in \mathbb{R}^{V \times D}$ is the weight embedding matrix, D is the hidden size, V is the entity size. As to Coke, we use cross-entropy between the one-hot label $\mathrm{y_1/y_n}$ and the prediction $\mathrm{P_1/P_n}$ as training loss:

$$Loss(X) = -\sum_t y_t \log p_t \tag{13}$$

where $\mathrm{y_t}$ and $\mathrm{p_t}$ are the t-th components of $\mathrm{y_1/y_2}$ and $\mathrm{P_1/P_2}$. As same as the model Coke, we use a label smoothing strategy to lessen this restriction.

5 Experiment

We measure our model's effectiveness of representations in link prediction. The subsequent section describe our complete experimental setup, including baselines, datasets, evaluation metrics and experimental analysis.

5.1 Datasets

We choose four widely used knowledge graph datasets and an external text corpus to train and evaluate the model, which are described as follows:

Knowledge Graph Dataset. The complete knowledge graph is very large, so its subset is generally used to evaluate the performance of knowledge graph representation learning method. We adopt four KG datasets, namely, FB15k, WN18, FB15K-237 and WN18RR. FB15k and WN18 are subsets of Freebase and WordNet respectively.The detailed statistical data is shown in Table 1.

Text Corpus. The text corpus is introduced in (Zhigang Wang et al., 2016), sampling from Wikipedia. This corpus consists mainly of unstructured natural language documents composed of natural language statements. Wang et al. deleted the external text corpus and improved the quality of the corpus. Finally, the pre-processed text corpus and entities on KG are linked, and the statistical data results are shown in Table 2.

Table 1. KG dataset statistical information in experiment.

Dataset	FB15k	WN18	FB15k-237	WN18RR
Entities	14951	40,493	14541	40493
Relations	1245	18	237	11
Train	483,142	141,442	272,115	86,835
Valid	50,000	5,000	17,535	3,034
Test	59,071	5,000	20,466	3,134

Table 2. Text corpus and KG entity alignment result statistics.

KG	Entities	Annotated entities	Word stems
WN18	40,943	32,249	1,529,251
FB15K	14,951	14,405	744,983

5.2 Evaluation Protocol

Link prediction task is a commonly used knowledge graph representation learning evaluation method. Given a correct triplet (h, r, t), the task of link prediction is to predict the missing h or t in the case of $(?, r, t)$ or $(h, r, ?)$. In the evaluation part, h or t in each triplet (h, r, t) in the test data set is replaced with **Mask**, and then this sequence is put into our model to obtain the distribution of all entities, and then conduct the descending order according to the predicted value, and finally get the rank of the predicted entity. In this paper, Mean Reciprocal Rank(MRR) and Hist@n are used as evaluation indexes of the model. MRR is the average of the reciprocal ranking of the correct entities in all test samples in the test triples. Hits@n means the proportion of the valid test triples ranking in top n predictions. Higher MRR or higher Hits@n indicate better performance.

5.3 Baslines

To verify the express ability of our model, We compare our model with several state-of-art methods, including method that in Euclidean Space(TransE [3], TransR [11]), method that in Complex Space(RotatE [18]), method that in Hyperbolic Space(HyperKG [7]) and method that use context information and external information (TEKE [22], Coke [20]).

- **TransE** is the first vector translation based model by assuming that the head entity embedding should be close to the tail entity embedding after relational embedding translation.
- **TransR** is a variant of TransE. TransR improves the modeling capability of complex relationships by embedding entities and relationships into different semantic Spaces through mapping matrices.
- **TEKE** uses textual description information of entities and context of relations to improve the effect of knowledge embedding.

– **RotatE** extends TransE's ideas by representing entities into complex space. RotatE regards relations as rotations from head entity to tail entity.
– **CoKE** proposes that the representation of entity and relation should represent dynamically depending on the context in which they are expressed. Triples are converted into sequential inputs through Transformer model.

Table 3. Link prediction results on FB15k and WN18 dataset.

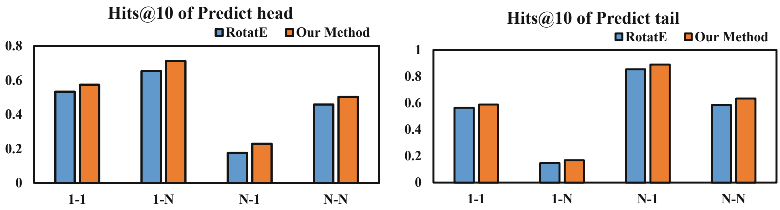
	FB15K				WN18			
Evaluation Protocol	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
Method that in Euclidean Space:								
TransE	0.523	0.476	0.528	0.572	0.541	0.513	0.549	0.615
TransR	0.556	0.484	0.544	0.605	0.548	0.52	0.569	0.642
TEKE	0.672	0.612	0.634	0.735	0.677	0.623	0.643	0.734
Coke	0.744	0.691	0.737	0.842	0.818	0.825	0.876	0.894
Method that in Complex Space:								
RotaE	0.712	0.708	0.714	0.732	0.757	0.73	0.759	0.787
Method that in Hyperbolic Space:								
HyperKG	0.687	0.656	0.693	0.634	0.739	0.724	0.735	0.766
Our method	0.807	0.732	0.785	0.867	0.824	0.838	0.887	0.906

5.4 Main Results

To verify the effectiveness of our model, we compare with several representative and widely used representation learning models. The experimental results of FB15K and WN18 are shown in Table 3. Part of the experimental results refer to the original paper, part is the result of self-tuning. Compared with the original paper of Baseline, there is a gap between our results and the Baseline, which may be due to data filtering. The optimal model in Baseline is CoKE. This illustrates the validity of considering that entities are dynamically generated depending on their context. On datasets FB15K and WN18, our model outperforms other models in terms of evaluation indicators. On FB15K, Hist@10 and MRR increased by 2.5% and 6.3%, respectively. This shows that it is useful to add more semantic and spatial considerations when doing dynamic contextual representations.

5.5 Complex Relation Study

Focus on the complex relations especially for the issue of 1-1, 1-N, N-1 and N-N, we conduct experiments on different relation categories. We choose the FB15k-237 as our datasets due to its abundant multi-relations and denser graph structure. We choose Rotate [18] as the baseline. From Fig. 2, we can find that our model is superior to the Baseline model in handling complex relationships. Compared with RotatE, our model has similar prediction performance in simple relationships 1-1 and 1-N. But in the case of complex relationships N-1and N-N, the predictive power of our model improved. Significantly. This indicates that considering contextual information is beneficial to improve the ability of our model to deal with complex relations.



(a) Hits@10 of Predict head (b) Hits@10 of Predict tail

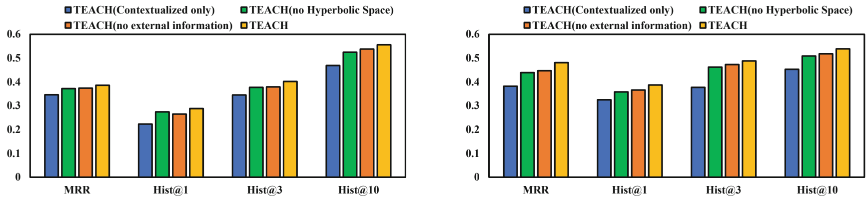
Fig. 2. Complex relation test on FB15k-237 dataset. Following the [21], the relations are divided into four categories: one-to-one (1-1), one-to-many (1-N), many-to-one (N-1), and many-to-many (N-N).

5.6 Ablation Study

We have given the comparisons and analysis about different models and our model has achieved a significant improvement over all baselines. In this section, in order to further analyze the enhancement effect of hyperbolic spatial information and descriptive text information on dynamically generated entity representation, we conducted an ablation experiment on the proposed model. We do the experiment on FB15k-237 and WN18RR dataset. Our model adds external semantic information and hyperbolic space embedded information modules on the basis of CoKE. In this section, we mainly analyze the improvement effect of external information embedding on the model.

Textual Embedding. In this section, we analyze the importance of external text and how the external semantic information affect the performance of our model. As we can see from Fig. 3. Compared with TEACH(Contextualized only), the performance of the model with external text improved. We can observe that, with the help of external text information, our model can improve input constraint. A possible reason is that the fusion of external text information can enhance the semantic information of the input. It will not lose semantic information when doing context training later.

Hyperbolic Translation. Compared with no hyperbolic spatial information, the effect of hyperbolic spatial information is obviously improved. We think it is the properties of hyperbolic space that capture the hidden information in the triplet and improve the representation ability. It can be seen that the effect of adding external text description information is better than adding hyperbolic spatial information only, but the two kinds of information do not completely coincide. Adding description information and spatial information at the same time can add constraints to the representation of dynamically generated entities and relations in context and make it more accurate.



(a) Ablation Study on FB15K-237 (b) Ablation Study on WN18RR

Fig. 3. Performance comparisons of information fusion model. Contextualized only; information fusion without hyperbolic space; Information fusion without external information.

6 Conclusion

Most of the existing knowledge graph representation models often ignore the context information of entities in the knowledge graph. This will lead to Knowledge Graph Representation Learning(KGRL) difficult to resolve complex relations. The existing Internet has a large amount of description text of entities, which helps to improve the ability of entity representation. In this work, We propose a new model, which considers the influence of external semantic information and different contexts on entity representation. We also consider the ability of hyperbolic space to capture hidden information, which makes it more suitable for knowledge representation learning tasks of knowledge graph. To fuse the external description information and the information captured in hyperbolic space, we propose a novel information fusion method for information aggregation. Experimental results show that adding external information constraints can improve the dynamic context representation of entities and relations.

As future work, we will simultaneously consider the loss in the process of knowledge fusion training and the loss in the process of context information training, because limit the fused information may not be well suited to downstream mission objectives. Finally, we will focus more on the coding of knowledge graph relational schema, including improving the coding capabilities of symmetric/antisymmetric, inverse, composite and subrelation schema.

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